

3.8.1 Alteration of the sequence of bases in DNA can alter the structure of proteins (A-level only)

Content	Opportunities for skills development
Gene mutations might arise during DNA replication. They include addition, deletion, substitution, inversion, duplication and translocation of bases. Gene mutations occur spontaneously. The mutation rate is increased by mutagenic agents. Mutations can result in a different amino acid sequence in the encoded polypeptide. • Some gene mutations change only one triplet code. Due to the degenerate nature of the genetic code, not all such mutations result in a change to the encoded amino acid. • Some gene mutations change the nature of all base triplets downstream from the mutation, ie result in a frame shift. Students should be able to relate the nature of a gene mutation to its effect on the encoded polypeptide.	